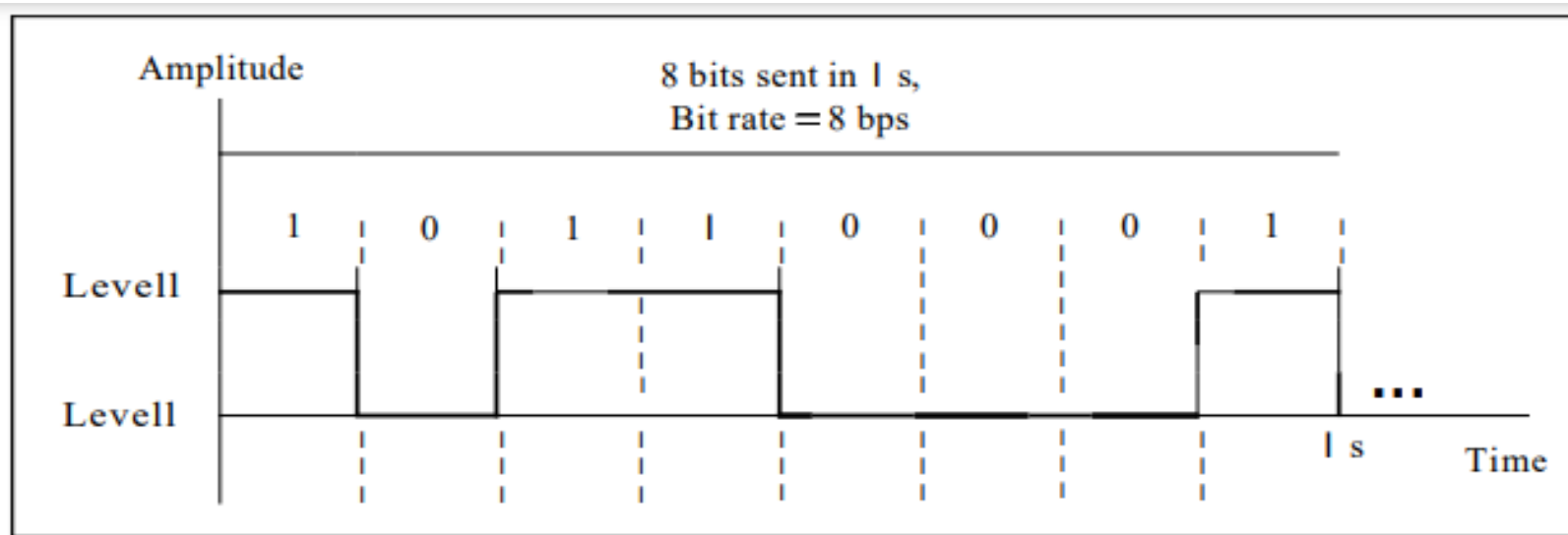
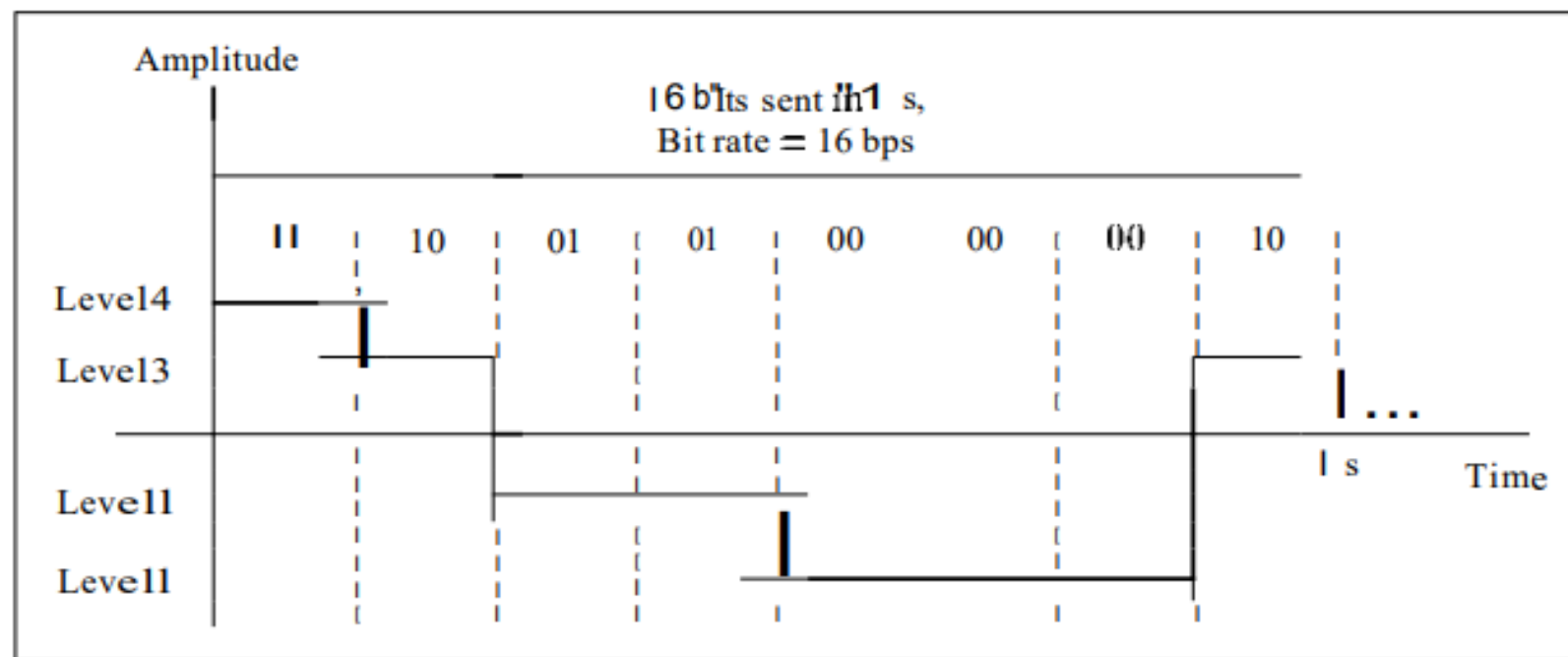


DIGITAL SIGNALS

- **0 & 1**
- **A digital signal can have more than two levels.**
- **A digital signal has eight levels. How many bits are needed per level?**
- **Number of bits per level = $\log_2 8 = 3$ Each signal level is represented by 3 bits.**



a. A digital signal with two levels



b. A digital signal with four levels

Bit Rate

Qus. Assume we need to download text documents at the rate of 100 pages per minute. What is the required bit rate of the channel? Given at a page is an average of 25 lines with 100 characters in each line. If we assume that one character requires 8 bits, the bit rate is

- $100 \times 25 \times 100 \times 8 = 2,000,000 \text{ bps} = 2 \text{ Mbps}$**

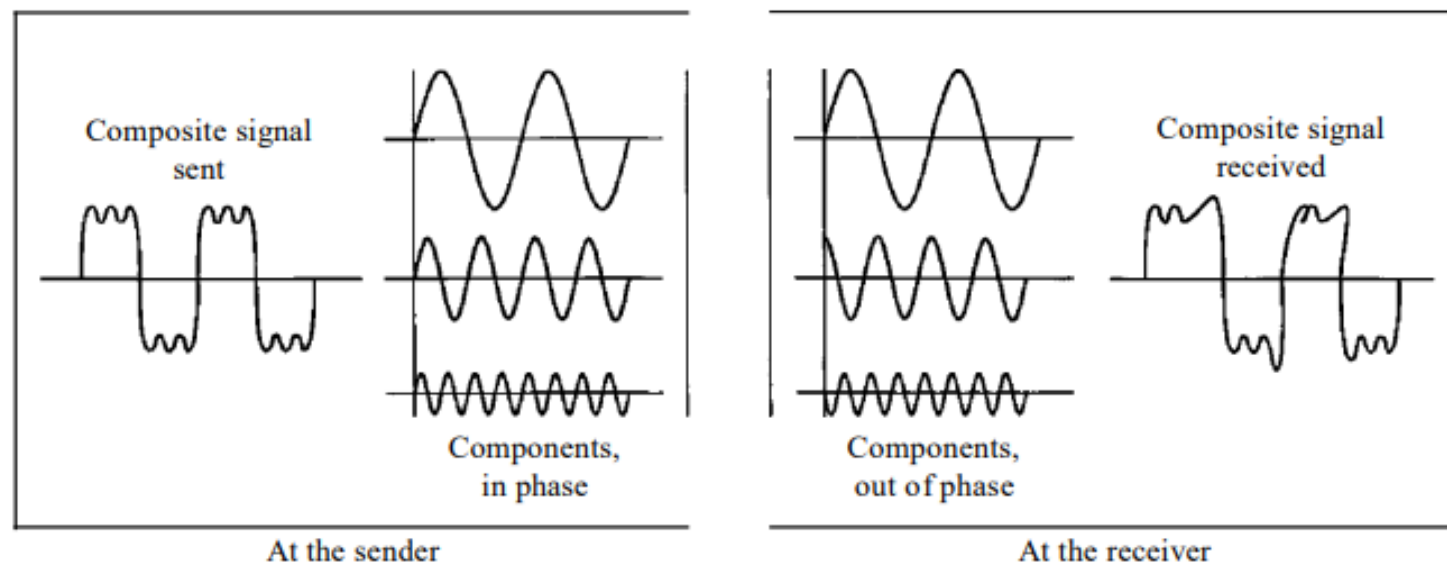
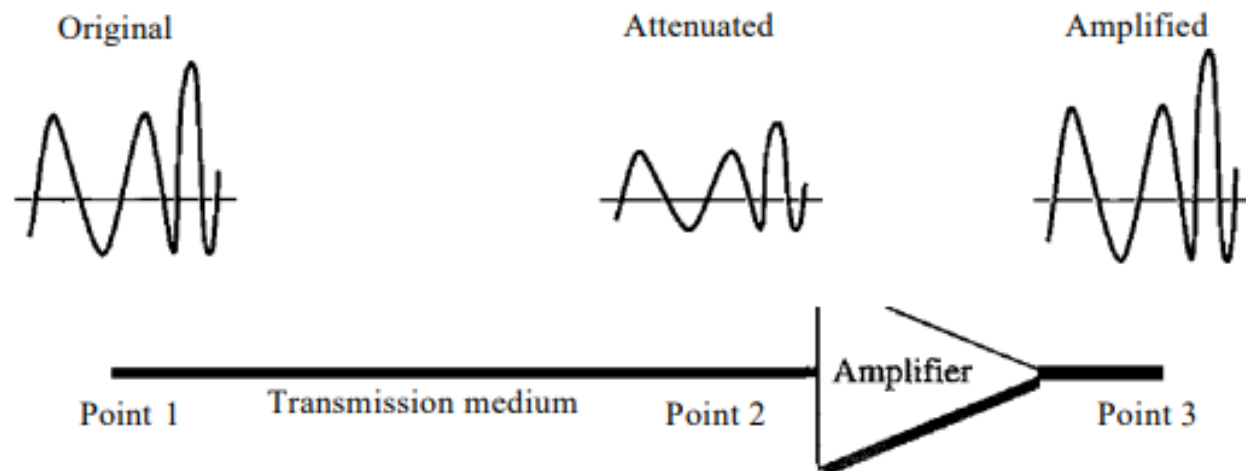
Transmission of Digital Signal

- **Baseband transmission or broadband transmission (using modulation).**
- **Baseband transmission means sending a digital signal over a channel without changing the digital signal to an analog signal.**
- **Broadband transmission or modulation means changing the digital signal to an analog signal for transmission.**

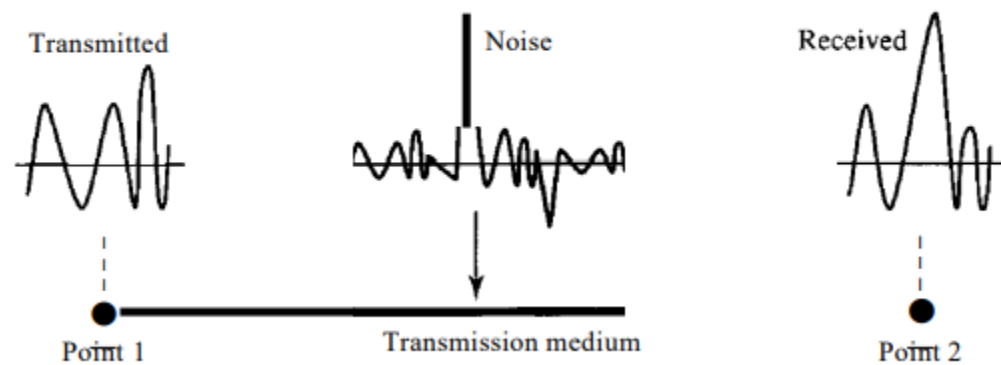
Causes of impairment

- **Attenuation means a loss of energy. When a signal, simple or composite, travels through a medium, it loses some of its energy in overcoming the resistance of the medium. That is why a wire carrying electric signals gets warm, if not hot, after a while.**
- **Some of the electrical energy in the signal is converted to heat. To compensate for this loss, amplifiers are used to amplify the signal.**
- **The decibel (dB) measures the relative strengths of two signals or one signal at two different points.**
- **Distortion means that the signal changes its form or shape.**
- **Noise is another cause of impairment. Several types of noise, such as thermal noise, induced noise, crosstalk, and impulse noise, may corrupt the signal.**

Attenuation



Noise



Bandwidth vs Throughput

- A network with bandwidth of 10 Mbps can pass only an average of 12,000 frames per minute with each frame carrying an average of 10,000 bits. What is the throughput of this network?
- Latency (Delay)= Propagation time +transmission time +queuing time + processing delay